

SEMESTER END EXAMINATIONS - MARCH 2024

Program	: B.E. - Common to CSE / ISE / AI & ML / AI & DS / CSE(CY) / CSE (AI & ML)	Semester	: III
Course Name	: Laplace Transforms and Vector Space	Max. Marks	: 100
Course Code	: CS / IS / AI / AD / CY / CI31	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Write the Laplace transform of a periodic function. CO1 (02)
 - Given $L\{\sin\sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, show that $L\left\{\frac{\cos\sqrt{t}}{\sqrt{t}}\right\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$. CO1 (04)
 - If $L\{f(t)\} = F(s)$ then prove that $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}\{F(s)\}$, where n is a positive integer. CO1 (07)
 - Evaluate the following integrals using Laplace transforms: CO1 (07)
 - $\int_0^{\infty} \frac{e^{-t} \sin t}{t} dt$
 - $L\{t \sin 3t \cos 2t\}$.
- Define Laplace transform of a function. CO1 (02)
 - Obtain the Laplace transform of $\left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^3$. CO1 (04)
 - Evaluate (i) $L\{te^{-4t} \sin 3t\}$ (ii) $L\left\{\frac{t - \sinh at}{t}\right\}$. CO1 (07)
 - A periodic function of period $\frac{2\pi}{\omega}$ is defined by CO1 (07)
$$f(t) = \begin{cases} E \sin(\omega t) & \text{for } 0 < t < \frac{\pi}{\omega} \\ 0 & \text{for } \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}, \text{ where } E \text{ and } \omega \text{ are positive constants, then show that } L\{f(t)\} = \frac{E\omega}{(s^2 + \omega^2)(1 - e^{-\pi s/\omega})}.$$

UNIT - II

- Define the Dirac-Delta function and sketch its graph. CO2 (02)
 - Evaluate $L^{-1}\left(\log\left(\frac{s^2 + 1}{s(s+1)}\right)\right)$. CO2 (04)
 - Express the function $f(t) = \begin{cases} t^2, & 0 < t < 2, \\ 4t, & 2 < t < 4, \\ 8, & t > 4. \end{cases}$ in terms of the Heaviside function and hence find its Laplace transform. CO2 (07)

- d) A voltage Ee^{-at} is applied at $t=0$ to a circuit of inductance L and resistance R . Show that the current at time t is $\frac{E}{R-al}(e^{-at} - e^{-Rt/L})$ using Laplace transforms. CO2 (07)
4. a) Define unit step function and represent graphically. CO2 (02)
- b) Find the inverse Laplace transform of $\frac{3s+2}{s^2-s-2}$. CO2 (04)
- c) Verify convolution theorem for the following pair of functions $f_1(t)=t$ and $f_2(t)=\cos t$. CO2 (07)
- d) Using Laplace transform, solve the simultaneous differential equations $\frac{dx}{dt} - 2y = \cos 2t$; $\frac{dy}{dt} + 2x = \sin 2t$, given $x(0)=1$, $y(0)=0$. CO2 (07)

UNIT - III

5. a) Determine whether the set $\left\{\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix}\right\}$ is a bases over the vector space M_{22} of set of all 2×2 matrices. CO3 (06)
- b) Show that the transformation $T: P_2 \rightarrow P_2$ defined as $T(ax^2 + bx + c) = (a+b)x^2 + c$ is linear. Find the image of $5x^2+6x+1$ under this transformation. CO3 (07)
- c) Show that a linear operator $T: R^2 \rightarrow R^2$ defined by $T(x) = Ax$ rotates a vector x through an angle θ about the origin if $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$. CO3 (07)
6. a) Determine the matrix that describes a reflection in the x -axis, followed by a rotation through $\frac{\pi}{2}$ followed by a contraction of factor $1/3$. Find the image of the point $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$ under this sequence of mappings. CO3 (06)
- b) Consider the bases $B = \{(1, 2), (3, -1)\}$ and $B' = \{(3, 1), (5, 2)\}$ of R^2 . Find the transition matrix from B to B' . If u is a vector such that $u_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, find $u_{B'}$. CO3 (07)
- c) Determine the kernel and range of the transformation defined by the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$. Hence verify rank-nullity theorem. CO3 (07)

UNIT- IV

7. a) Find the complete solution of $x+3y+3z=1$, $2x+6y+9z=5$, $-x-3y+3z=5$. CO4 (06)
- b) Let $y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $u = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$, find the orthogonal projection of y onto u . CO4 (07)
- Write y as the sum of two orthogonal vectors, one in $\text{span}\{u\}$ and one orthogonal to u .

- c) Find the dimension and basis for four fundamental subspaces of CO4 (07)

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}.$$

8. a) Find QR factorization of the matrix $A = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$. CO4 (10)

- b) Find the least-square solution of $AX = b$ for CO4 (10)

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix} \text{ and } b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}.$$

UNIT - V

9. a) Express the quadratic form $Q(x) = x_1^2 - 8x_1x_2 - 5x_2^2$ as quadratic forms with no-cross product terms. CO5 (05)

- b) Find the steady state solution of the Markov matrix $A = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$ CO5 (05)

- c) Find the singular value decomposition(SVD) of the matrix $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$. CO5 (10)

10. a) Check whether the following matrix is positive definite or not CO5 (05)

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}.$$

- b) Determine whether or not the matrix $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ is Unitary. CO5 (05)

- c) Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$. CO5 (10)
